



Circular economy meets water treatment: one-pot synthesis of agricultural waste-based bi-functionalized ordered mesoporous silica for tetracycline removal via systematic adsorption studies

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Abstract

This research endeavors to elucidate the process of developing and characterizing a novel plant-derived biogenic-based bi-functionalized MCM-41 (Mobil Composition of Matter No. 41), abbreviated as Bif-MCM-41, prepared through an environmentally friendly one-pot method. The Bif-MCM-41 material was functionalized with two different silane coupling agents with different functional groups to enhance its adsorption properties. XRD and TEM confirmed the mesoporous structure constructed from well-ordered hexagonal arrays of parallel microchannels, while FESEM revealed uniform Bacillus-like (rod-shaped) morphology. The material exhibited a high surface area ($988 \text{ m}^2 \text{ g}^{-1}$), a pore volume of $0.72 \text{ cm}^3 \text{ g}^{-1}$ and a bimodal pore distribution (1.3 nm and 2.5 nm) based on 2D-NLDFT method, confirming a well-developed micro-mesoporous structure. Adsorption studies for tetracycline were conducted under varying conditions, and both linear and nonlinear isotherm and kinetic models were applied to assess adsorption behavior. The maximum Langmuir adsorption capacity reached 765.4 mg g^{-1} at 293 K. Nonlinear fitting provided a more accurate representation of adsorption behavior, and kinetic studies indicated a pseudo-first-order mechanism, suggesting surface reaction dominance. Thermodynamic analysis confirmed the process as spontaneous and exothermic, driven by π - π stacking and hydrogen bonding. The combination of biogenic synthesis, dual functionalization, and outstanding adsorption performance represents a unique contribution to the development of sustainable adsorbents. This study offers both mechanistic insight and practical relevance, bridging green chemistry with high-efficiency pollutant removal.

Keywords Adsorption · Bi-functionalized MCM-41 · Phytogenic silica adsorbent · One-pot synthesis · Tetracycline removal

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List of symbols

$[\text{adsorbate}]^{\circ}$	The unitary standard concentration of the adsorbate (1 mol L^{-1})
2D-NLDFT	The two-dimensional version of the non-local density functional theory
α	The initial adsorption rate in Elovich equation ($\text{mg g}^{-1} \text{ min}^{-1}$)
α_{RP}	The RP isotherm constant (L mg^{-1})
β	The desorption constant (g mg^{-1})
γ	The coefficient of activity (dimensionless)
$\Delta H^{\circ}_{\text{Ads.}}$	The enthalpy change of adsorption (kJ mol^{-1})
$\Delta S^{\circ}_{\text{Ads.}}$	The entropy change of adsorption ($\text{kJ K}^{-1} \text{ mol}^{-1}$)
$\Delta G^{\circ}_{\text{Ads.}}$	Gibbs free energy of adsorption (kJ mol^{-1})

AEAE-APTMS	3-[2-(2-Aminoethylamino)ethylamino]propyltrimethoxysilane
BDPP-APTES	3-[Bis[(diphenylphosphino)methyl]amino]propyltriethoxysilane
B_T	The Temkin isotherm constant (J mol^{-1})
BET	Brunauer–Emmett–Teller
BJH	Barrett, Joyner, and Halenda
Bif-MCM-41	Bi-functionalized MCM-41
C_e	Equilibrium concentration of adsorbate (mg L^{-1})
C_{IPD}	The boundary layer thickness in IPD equation (mg g^{-1})
C_i	Initial concentration of adsorbate (mg L^{-1})
C_t	Concentration at time t (mg L^{-1})
CTAB	Cetyltrimethylammonium bromide
FESEM	Field emission scanning electron microscopy
FTIR	Fourier transform infrared
g	The exponent in RP model (dimensionless)
k_1	The rate constant of PFO (min^{-1})
k_2	The rate constant of PSO ($\text{g mg}^{-1} \text{min}^{-1}$)
k_{diff}	The IPD rate constant ($\text{mg g}^{-1} \text{min}^{-0.5}$)
K_e^o	Equilibrium constant (dimensionless)
K_F	Freundlich constant ($(\text{mg g}^{-1})^{1/n}$)
K_L	The Langmuir constant at given temperatures (L mg^{-1})
K_{RP}	The RP constant (L g^{-1})
K_T	The Temkin isotherm constant (L mg^{-1})
MCM-41	Mobil composition of matter no. 41
MW	The molecular weight of the adsorbent (g mol^{-1})
n	The heterogeneity factor in Elovich equation (unitless)
Na_2SiO_3	Sodium silicate
NSMs	Nanoporous silica materials
PFO	Pseudo-first-order
PSO	Pseudo-second-order
R	The universal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)
RP	Redlich–Peterson
Q_e	Adsorption capacity at equilibrium (mg g^{-1})
$Q_{e,\text{cal}}$	The calculated equilibrium adsorption capacity (mg g^{-1})
Q_m	Maximum adsorption capacity (mg g^{-1})
$Q_{m,\text{exp}}$	Experimental maximum adsorption capacity (mg g^{-1})
$Q_{m,\text{cal}}$	Maximum adsorption capacity estimated from the Langmuir model (mg g^{-1})
Q_t	Adsorption capacity at time t (mg g^{-1})

T	Temperature (K)
TEM	Transmission electron microscopy
t	Time (min)
V	Solution volume (L or mL)
W	Adsorbent dose (g or mg)

Introduction

The burgeoning global demand for sustainable water treatment solutions has led to an increasing interest in developing efficient techniques for removing pollutants. Various techniques such as membrane filtration, advanced oxidation processes, chemical precipitation, and biological treatment have been explored for the removal of emerging pollutants from water (Eniola et al. 2022). However, many of these methods suffer from limitations such as high operational cost, limited efficiency for trace-level pollutants, and generation of secondary waste. In contrast, adsorption stands out as a cost-effective, efficient, and scalable method with the ability to remove a wide range of contaminants, including antibiotics, even at low concentrations. Its simplicity, flexibility in adsorbent design, and ease of regeneration make it a highly attractive choice for sustainable water treatment applications (Rani et al. 2020; Rashid et al. 2021). Unlike chemical treatments, which often introduce secondary pollutants, and physical methods, such as filtration, which are limited by particle size, adsorption offers a highly versatile, efficient, and eco-friendly approach (Ukhurebor et al. 2024). It can target an extensive array of pollutants, encompassing various polluting substances such as organic compounds, heavy metals, and emerging contaminants like pharmaceuticals, with high selectivity and reusability of adsorbents. Adsorption processes, particularly those using advanced materials, also benefit from lower energy requirements and operational simplicity, making them particularly attractive for decentralized and rural water treatment systems (Sahoo and Prelot 2020). As global water scarcity and pollution intensify, the proverb "necessity is the mother of invention" aptly captures the impetus behind the innovation and adoption of adsorption technique in water treatment.

The escalating presence of pharmaceutical residues in aquatic environments poses a significant challenge, with tetracycline, a widely used antibiotic, emerging as a contaminant of critical concern (Scaria et al. 2021). Tetracycline is extensively used in human and veterinary medicine, and its overuse has led to its pervasive occurrence in water bodies, primarily through wastewater discharge. The persistence of tetracycline in the environment is alarming due to its potential to induce antibiotic resistance, disrupt aquatic ecosystems, and pose health risks to humans (Leichtweis et al. 2022). Tetracycline contamination in the environment primarily arises from its high stability and limited metabolic

breakdown in humans and animals (Amangelsin et al. 2023). This antibiotic is not completely absorbed or metabolized within the body, leading to its release into ecosystems (Xu et al. 2021a). Exposure to tetracycline adversely impacts algal populations, suppressing the growth of various algae species at concentrations between 0.25 and 30 mg L⁻¹ (Xu et al. 2019). Additionally, tetracycline exerts dose-dependent effects on both phytoplankton and zooplankton communities (Xu et al. 2021b). Given its recalcitrant nature and the potential for bioaccumulation, the removal of tetracycline from water supplies is not just desirable but essential. Thus, there is an urgent need to develop sustainable, efficient, and cost-effective methods to reduce the levels of tetracycline and other pharmaceuticals in water, aligning with the broader global goals of environmental protection and public health.

In recent years, nanomaterials derived from agricultural wastes have emerged as a promising solution in the quest for sustainable environmental applications, particularly in water treatment (Shirazian et al. 2025a). Among these, nanoporous silica materials (NSMs) stand out as a well-established family, characterized by their chemical stability, tunable pore sizes, biodegradable nature, and high surface area (Yang et al. 2024). These materials can be derived from the silica content in agricultural by-products, such as rice husk, sugarcane bagasse, sorghum, and sedge (*Carex riparia*), which are not only abundant and non-toxic but also biodegradable (Sarkar et al. 2021; Alsaab et al. 2024; Shirazian et al. 2025a). Sedge, in particular, presents a promising and sustainable source of silica due to several favorable characteristics. It is widely available in wetland ecosystems, typically grows in abundance without requiring intensive agricultural inputs, and holds minimal economic value as a commercial crop. These factors make it an ideal candidate for resource recovery without disrupting food or industrial supply chains. Moreover, sedge biomass can be readily processed to extract high-purity phytogenic silica or plant-derived biogenic silica through environmentally benign methods, offering a practical pathway toward the synthesis of silica-based materials. The valorization of sedge not only supports waste minimization and circular economy principles but also provides a green and cost-effective approach for producing high-value functional materials for environmental remediation, particularly in water treatment applications (Soltani et al. 2022).

To enhance the physicochemical properties of NSMs, particularly their adsorption efficiency, surface functionalization plays a pivotal role. This can be achieved through two general approaches: the one-pot or direct method and the post-synthesis or post-grafting method (Cashin et al. 2018). The one-pot method involves the simultaneous synthesis and functionalization of the material, allowing for a more uniform distribution of functional groups within the silica matrix. This method is particularly advantageous for adsorption applications due to its ability to create tailored

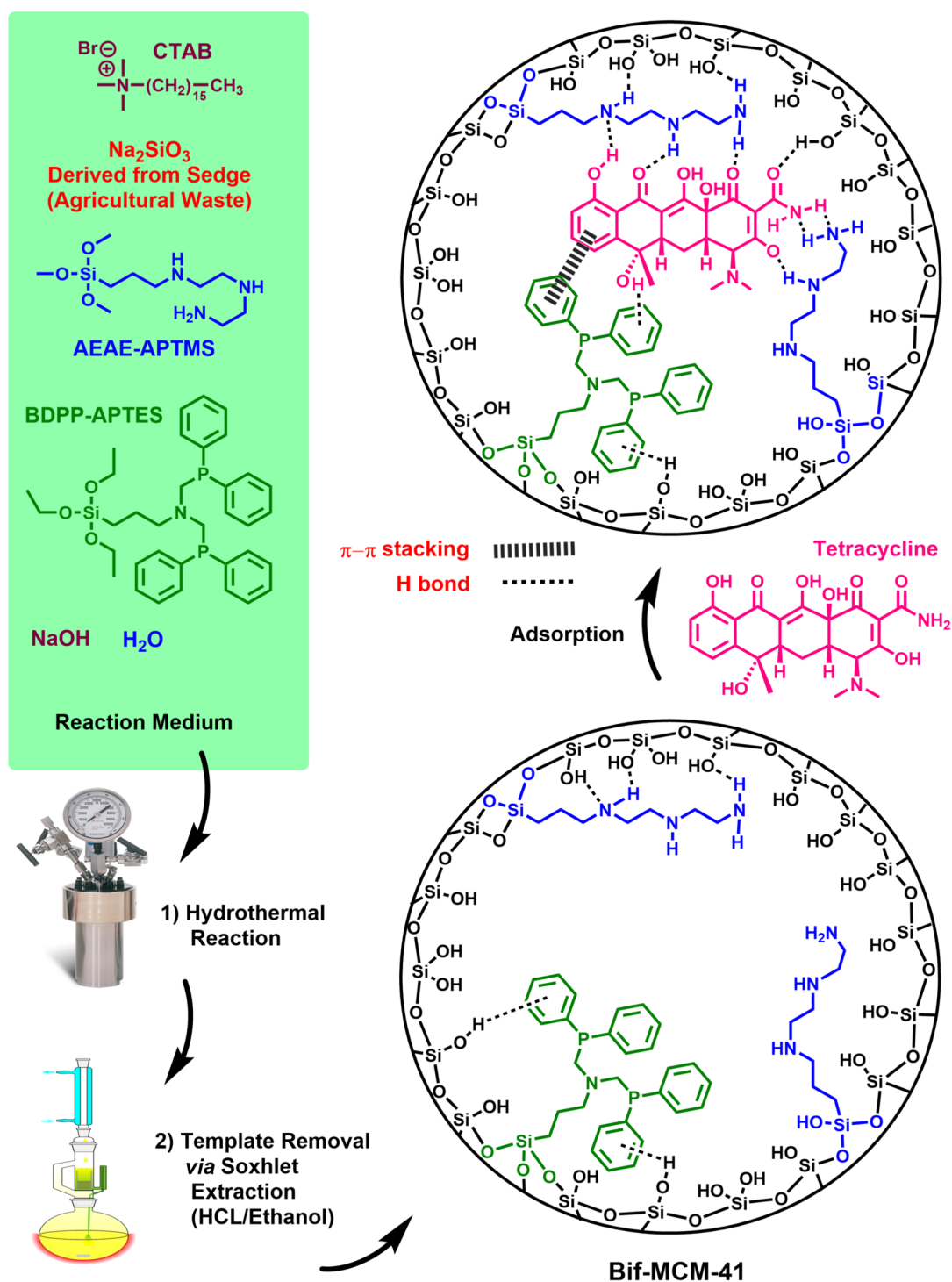
adsorbents with high selectivity and capacity (Soltani et al. 2022; Shirazian et al. 2024). On the other hand, the post-grafting method involves the modification of pre-formed silica structures, offering greater flexibility in introducing specific functional groups but often at the cost of less uniformity and potential pore blockage (Soltani et al. 2021). The choice between these methods depends on the intended application, with the one-pot method often preferred for adsorption due to its simplicity and efficiency in producing materials with high-performance characteristics.

In this study, we report a green and sustainable approach for the preparation of bi-functional mesoporous silica, specifically MCM-41 functionalized with amine and phenyl groups, referred to as bi-functionalized-MCM-41 (Bif-MCM-41, bi-functionalized = co-functionalization with amine and phenyl groups via two different silane coupling agents) (Scheme 1). Two silane coupling agents, namely 3-[Bis[(diphenylphosphino)methyl]amino]propyltriethoxysilane (BDPP-APTES), bearing phenyl groups, and 3-[2-(2-aminoethylamino)ethylamino]propyltrimethoxysilane (AEAE-APTMS), bearing amine groups, were used to functionalize the surface of the Bif-MCM-41 in one-pot functionalization strategy. This novel material was synthesized using a one-pot, hazardous solvent-free method and characterized using various instrumental techniques, including powder XRD, FTIR, N₂ nitrogen adsorption–desorption isotherms, FESEM, and TEM. The Bif-MCM-41 was then applied for the adsorption of tetracycline from water, and a comprehensive set of adsorption studies was conducted to determine the optimal conditions for maximum removal efficiency. The results highlight the potential of Bif-MCM-41 as an effective adsorbent for tetracycline, providing a promising solution for the sustainable treatment of pharmaceutical-contaminated water, demonstrating that “necessity is the mother of sustainable innovation”.

Material and methods

To streamline the article and emphasize the results and discussions, several sections have been condensed and moved to the Supplementary Material. These sections include:

1. Materials, Instruments, and Methods (Section S1)
2. Synthesis of Materials, covering Silica Source Preparation and the Synthesis of Bif-MCM-41 (Section S2)
3. Adsorption studies, encompassing the investigation of pH and adsorbent dose effects, simultaneous effect of initial concentration and temperature, effect of contact time, regeneration/reusability test, and adsorption equations (Section S3)
4. Additional Figures (Section S4)
5. Supplemental Tables (Section S5)



Scheme 1 Schematic representation of the overall synthesis procedure for Bif-MCM-41, involving the functionalization of mesoporous silica MCM-41 with amine and phenyl groups derived from sedge

(*Carex riparia*) waste. This bi-functional material is designed for the efficient adsorption of tetracycline from water

By consolidating these sections, the focus is placed on the crucial findings and interpretations while still providing comprehensive information on the study's methodology and supporting data.

Synthesis of Bif-MCM-41

A brief synthesis method for Bif-MCM-41 is provided here; full details are available in the Supplementary Material,

Section S2. Silica powder extracted from sedge was dispersed in aqueous NaOH under ultrasonication and then refluxed at controlled temperature. A CTAB solution was added gradually under stirring, and the pH was adjusted to 11. A mixture of AEAE-APTMS and BDPP-APTES (2:1 molar ratio) was introduced sequentially. After vigorous stirring, the mixture was hydrothermally treated at 373 K for 72 h in a Teflon-lined autoclave, aged further, and then filtered, washed, and vacuum-dried. The final material underwent Soxhlet extraction using acidic ethanol to remove the surfactant template.

Results and discussion

Synthesis of Bif-MCM-41

The synthesis of Bif-MCM-41 involved the strategic utilization of various components, each playing a pivotal role in the process. Na_2SiO_3 derived from plant-derived biogenic silica or phytogenic silica extracted from sedge served as the primary precursor for the silica framework, providing an eco-friendly source of mesoporous Bif-MCM-41 silica. CTAB, a cationic surfactant, acted as the templating agent, directing the formation of the characteristic hexagonal array of mesochannels in the MCM-41 structure through the self-assembly of its molecules in the solution. Water was employed as the solvent, facilitating the dissolution and interaction of the various components throughout the synthesis process.

Ghorbani et al. (2013) reported that the silica content in sedge ash varies depending on the treatment method. Without acid treatment, the silica content was 31.42% under uncontrolled combustion conditions and 33.14% at 600 °C. When acid treatment was applied, the purity significantly increased—reaching 93.82%, 87.25%, and 98.05% using reflux, leaching, and a combined leaching–reflux method, respectively. Similarly, Soltani and co-workers (2022; Alshehri et al. 2025) reported a high silica purity of approximately 98% using a sequential process involving leaching, refluxing, and combustion. These findings support our choice of sedge as a viable, sustainable, and high-purity precursor for the synthesis of ordered mesoporous silica in this study.

To tailor the surface adsorption properties of Bif-MCM-41 and enhance its adsorption affinity for tetracycline, two functionalization agents were utilized: AEAE-APTMS and BDPP-APTES. These agents anchored phenyl and amine groups to the pores of the adsorbent, promoting the establishment of π – π stacking interactions (mainly between phenyl rings of BDPP-APTES and phenyl ring of tetracycline) and hydrogen bonding (between heteroatoms on the adsorbent surface and hydroxyl, ketone, and amine groups of tetracycline) as shown in Scheme 1. Additionally,

London dispersion forces between the adsorbent and adsorbate further enhance the overall interaction (not depicted in Scheme 1). This functionalization not only increased the adsorption capacity of the material but also contributed to its selectivity for tetracycline removal.

The synthesis of Bif-MCM-41 progressed through the formation of cylindrical micelles by the self-assembly of CTAB molecules in the presence of Na_2SiO_3 . The silicate network was then formed around these micelles, with the functionalization agents integrated into the structure. Hydrothermal conditions played a crucial role in pore formation and stability, as they facilitated the proper organization and condensation of the silica framework, ultimately resulting in the characteristic hexagonal mesoporous structure of MCM-41.

Instrumental techniques for the characterization of Bif-KCC-1

XRD patterns of the Bif-MCM-41 (Fig. 1a) exhibit three characteristic peaks at 2θ values of 2.2° , 3.9° , and 4.5° , which correspond to the (100), (110), and (200) crystallographic planes, respectively. These peaks are indicative of the well-ordered hexagonal mesoporous structure typical of MCM-41 materials. The (100) peak, appearing at the lowest angle (2.2°), is particularly intense and sharp, reflecting the high degree of structural order and large d-spacing, which is a hallmark of the MCM-41 framework. The presence of the (110) and (200) peaks, albeit with lower intensity, further confirms the long-range order within the mesoporous channels (Soltani et al. 2019a, b). Functionalization of MCM-41 with organic moieties may cause a slight shift in the 2θ values or a reduction in peak intensity, which is attributed to the alteration in pore wall thickness and the introduction of guest molecules within the mesopores.

The FTIR spectrum of Bif-MCM-41 given in Fig. 1b reveals distinctive absorption bands linked to both the silica framework and the grafted organic groups. A broad band around $3400\text{--}3500\text{ cm}^{-1}$, indicating O–H stretching, suggests the presence of surface silanol groups (Si–OH) and adsorbed water. Si–O–Si asymmetric stretching vibrations, characteristic of the MCM-41 silica network, are evident between 1100 and 1000 cm^{-1} , with additional Si–O–Si symmetric stretching at 800 cm^{-1} and bending at 460 cm^{-1} , confirming the formation of silica structure (Soltani et al. 2019b, 2020).

The presence of absorption bands due to the introduction of AEAE-APTMS and BDPP-APTES confirms the successful attachment of organic moieties via one-pot grafting silane coupling agents on the surface of the silica network. Absorption bands around $3000\text{--}3100\text{ cm}^{-1}$ correspond to C–H stretching in phenyl rings of BDPP-APTES. C–H stretching vibrations from methylene groups ($-\text{CH}_2-$) are observed

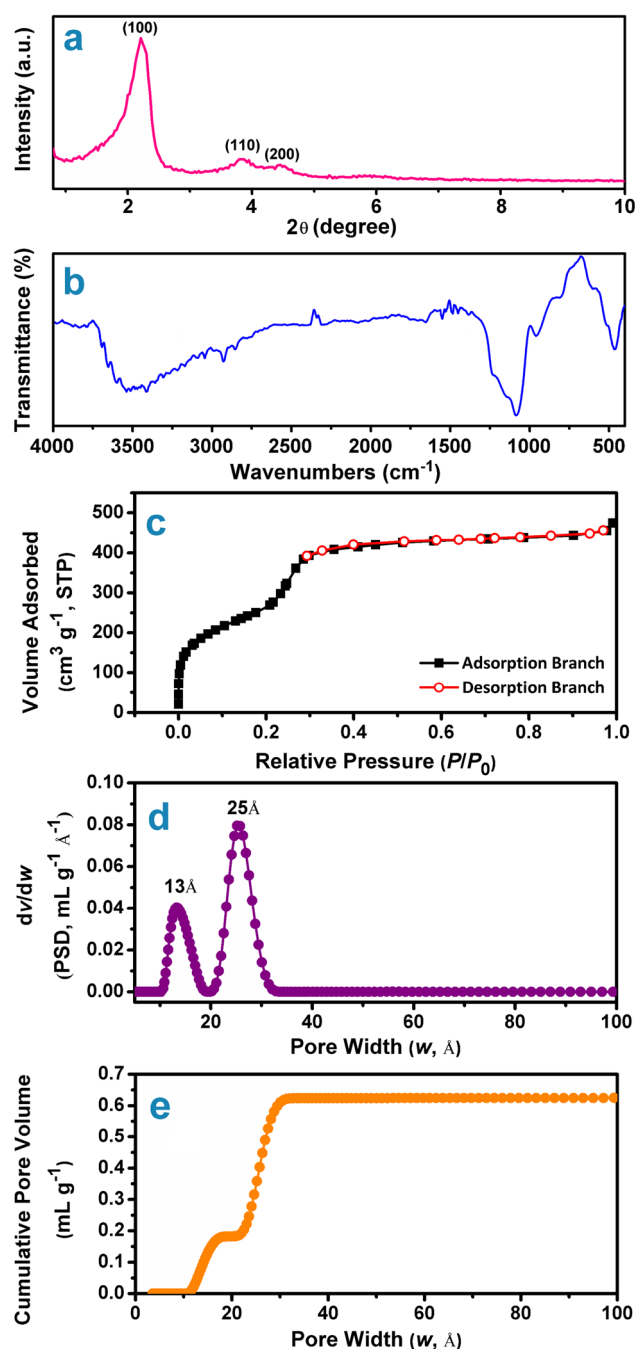


Fig. 1 **a** The low-angle XRD pattern of Bif-MCM-41. **b** The FTIR spectrum of the Bif-MCM-41. **c** N₂ adsorption–desorption isotherms of Bif-MCM-41 at 77 K, showing a Type IV isotherm with a narrow hysteresis loop. **d** 2D-NLDFT analysis indicating bimodal pore size distribution with peaks at 1.3 nm and 2.5 nm. **e** Cumulative pore volume distribution across different pore sizes

at around 2925 cm⁻¹ and 2850 cm⁻¹, while N–H stretching between 3200–3400 cm⁻¹ and bending near 1600–1500 cm⁻¹ confirm the presence of primary and secondary amines from AEAE-APTMS (Soltani et al. 2019a). These amine groups, capable of hydrogen bonding, are essential for enhancing the

material's adsorption capabilities. The presence of phenyl groups from BDPP-APTES is indicated by C=C stretching vibrations in the 1450–1600 cm⁻¹ range, and P–C stretching vibrations around 700–750 cm⁻¹ confirm the presence of phosphine groups (–PPh₂) (Li et al. 2006). Consistent with literature reports, these FTIR findings demonstrate successful surface modification of MCM-41 with AEAE-APTMS and BDPP-APTES, as evidenced by the appearance of new bands related to organic groups alongside retained silica framework vibrations, thus preserving the mesoporous structure while enhancing functional properties.

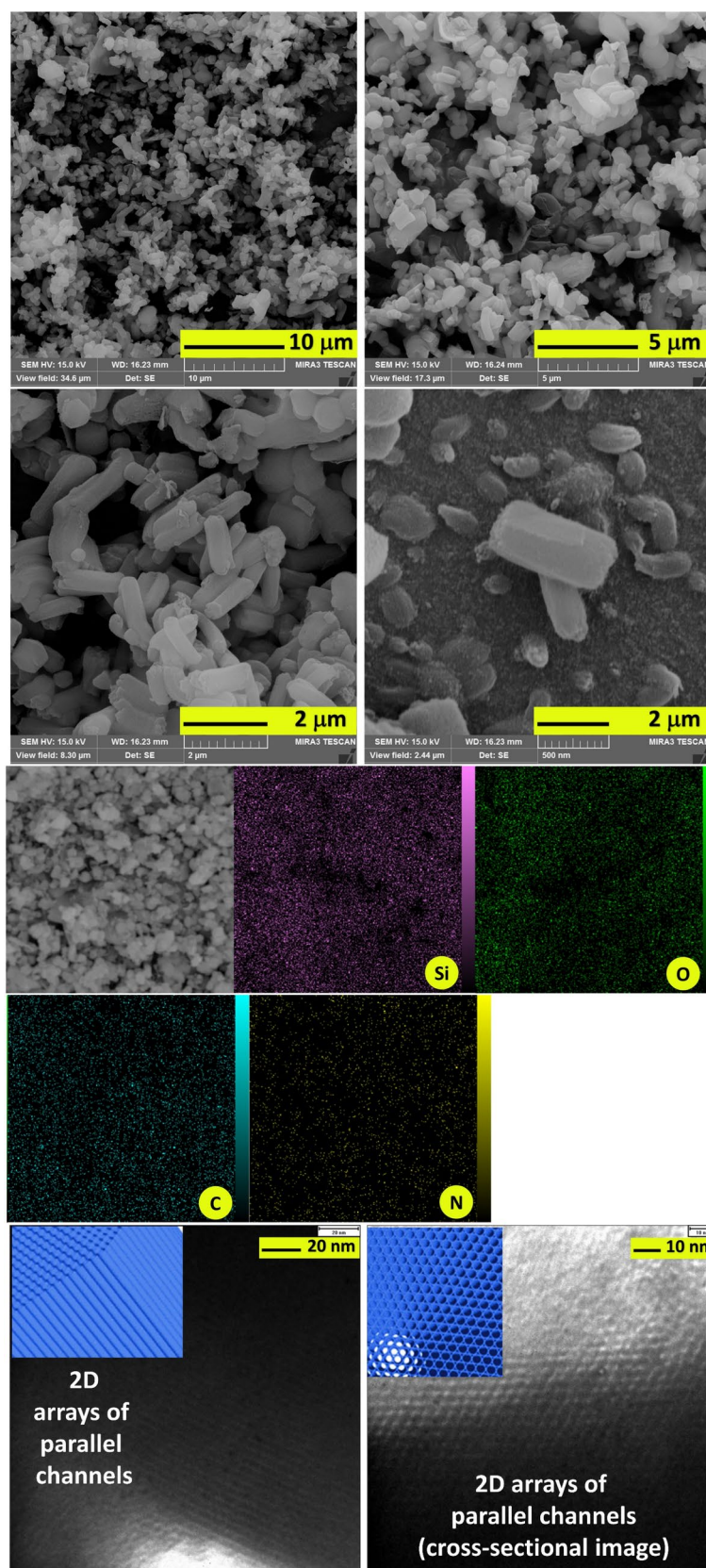
The N₂ adsorption–desorption isotherms of Bif-MCM-41 (Fig. 1c), prepared via the one-pot method, exhibit a Type IV isotherm with a narrow hysteresis loop, indicative of a structure with significant microporosity and some mesoporous features (Thommes et al. 2015). According to the BET model, the material possesses a surface area of 988 m² g⁻¹, a total pore volume of 0.72 cm³ g⁻¹, and a mean pore diameter of 2.7 nm. The BJH analysis reveals a pore volume of 0.71 cm³ g⁻¹ and an average pore diameter of 1.3 nm, emphasizing the presence of smaller mesopores. The t-plot method indicates a surface area slightly exceeding 1000 m² g⁻¹, confirming that a substantial portion of the pore distribution is within the microporous range, which correlates with the observed narrow hysteresis loop.

Further validation by 2D-NLDFT analysis (Fig. 1d, e) reveals two dominant peaks at 1.3 nm and 2.5 nm, suggesting a bimodal micro-mesoporous structure with a significant microporous character. We hypothesized that the presence of bimodal pore sizes in Bif-MCM-41 provides synergistic advantages. The smaller pores (~1.3 nm) are effective for size exclusion and offer strong confinement effects that can enhance interaction between the adsorbent and adsorbate (e.g., π – π stacking and hydrogen bonding). The larger pores (~2.5 nm) facilitate faster mass transport and diffusion of tetracycline molecules into the material. This hierarchical pore system ensures both high accessibility and strong adsorption affinity, which together contribute to the exceptional adsorption capacity observed.

The one-pot synthesis method, by facilitating concurrent functionalization and pore formation, enhances the stability and uniformity of the material, making it particularly suitable for applications requiring high adsorption capacity and selective separation. This method ensures that the functional groups are uniformly distributed, preserving the integrity of the pore structure and contributing to the material's performance in various adsorption and catalytic processes.

Four FESEM images of Bif-MCM-41 were captured at different magnifications (with scale bars at 10 μ m, 5 μ m, 2 μ m, and 500 nm) and are presented in Fig. 2, first and second rows. These images reveal a uniform particle morphology, indicative of the high degree of control of the functionalization process achieved during the one-pot synthesis

Fig. 2 FESEM images (first and second rows), FESEM mapping images (third and fourth rows), and TEM micrographs of Bif-MCM-41 (fifth row)



process. The uniformity in particle shape and size distribution suggests that the functionalization process—integrating AEAE-APTMS and BDPP-APTES—occurred homogeneously throughout the silica matrix, without inducing significant structural disruption. The observed Bacillus-like or rod-shaped morphology is characteristic of MCM-41 materials, reflecting the retention of the hexagonal mesoporous structure post-synthesis. This method ensures that the functional groups are uniformly distributed within the material, which is crucial for maintaining the material's surface area and porosity, essential for its intended applications.

Elemental mapping via FESEM mapping reveals a homogenous distribution of silicon (Si), oxygen (O), carbon (C), and nitrogen (N) throughout the modified MCM-41 structure (Fig. 2, third and fourth rows). The detection of Si and O is consistent with the silica framework of MCM-41, while the presence of C and N elements confirms the successful functionalization with organic moieties or nitrogen-containing compounds. The uniform distribution of these elements suggests that the functionalization process was highly effective, leading to an even modification of the MCM-41 surface, with no significant agglomeration or phase separation observed, which is critical for ensuring consistent material performance.

TEM analysis provides a more detailed view of the structural integrity and arrangement of the Bif-MCM-41. The TEM images (Fig. 2, fifth row) display well-ordered hexagonal arrays of parallel microchannels, which are emblematic of the MCM-41 mesostructure. The cross-sectional TEM images further reveal a two-dimensional array of these channels, confirming that the ordered mesoporous structure remains intact following one-pot functionalization.

This preservation of the hexagonal mesopores arrangement is crucial, as it ensures the material retains its high surface area and pore volume, both of which are essential for adsorption and catalytic applications.

Adsorption studies

To optimize the adsorption capacity, various factors will be systematically investigated in the following sections to achieve the maximum adsorption efficiency of the adsorbent.

The influence of adsorbent dose and pH

The adsorbent dose and solution pH are critical factors influencing the adsorption performance of Bif-MCM-41 for tetracycline removal. Detailed trends are illustrated in Fig. S1. As the adsorbent dose increased from 2.0 to 10.0 mg, the removal efficiency of tetracycline improved significantly, primarily due to the greater number of accessible active sites on the surface of the adsorbent. Specifically, when the pH was increased from 2.0 to 5.0, the tetracycline removal rose

markedly—from approximately 10%, 16%, and 19% at pH 2.0 to about 85%, 91%, and 93% at pH 5.0 for doses of 2, 5, and 10 mg, respectively. This enhancement is attributed to reduced competition between H^+ ions and tetracycline molecules for surface sites and to improved interaction between tetracycline and the functional groups on Bif-MCM-41.

At low pH values (e.g., pH 2.0), the surface of Bif-MCM-41 is highly protonated, particularly the amine and silanol groups, leading to electrostatic repulsion with the predominantly cationic tetracycline species and limited hydrogen bonding. As pH increases toward 5.0, deprotonation of silanol and phenol groups occurs gradually, enhancing hydrogen bonding and possibly π – π interactions between the aromatic rings of tetracycline and the phenol moieties. The amine groups (with $pK_a \sim 9$ –10) remain partially protonated, allowing favorable electrostatic interactions with zwitterionic or negatively charged TC species.

Between pH 5.0 and 7.0, a slower increase in removal efficiency is observed, reaching approximately 92%, 98%, and 99% at pH 7.0 for doses of 2, 5, and 10 mg, respectively. At this near-neutral pH, tetracycline exists mainly in its zwitterionic form, while the adsorbent surface exhibits a balanced distribution of neutral and deprotonated sites. This condition supports optimal hydrogen bonding, π – π stacking, and limited electrostatic attraction, contributing to high uptake.

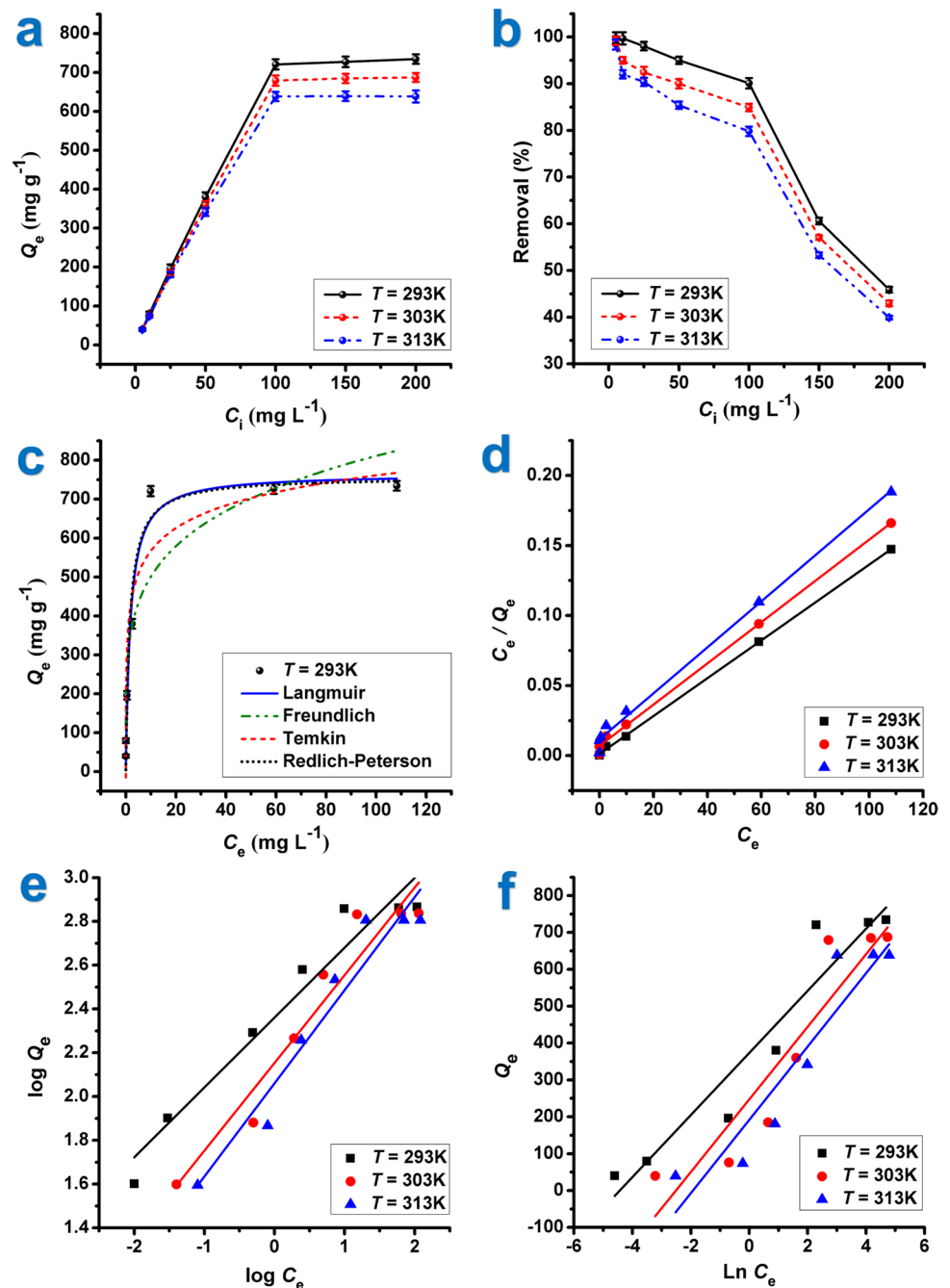
Beyond pH 7.0, the removal efficiency declines moderately (to $\sim 70\%$, 80% , and 81% at pH 10 for 2, 5, and 10 mg, respectively), likely due to increasing deprotonation of both the tetracycline molecules and the adsorbent's functional groups. At basic pH, electrostatic repulsion becomes more pronounced, and reduced hydrogen bonding capability leads to a decline in adsorption performance.

Moreover, although increasing the adsorbent dose from 5.0 to 10.0 mg results in slightly higher removal percentages, the incremental gain is minimal. This marginal improvement suggests near-saturation of available TC molecules at higher doses. Therefore, an adsorbent dose of 5.0 mg at pH 7.0 was determined to be optimal, providing high removal efficiency while maintaining material economy.

The influence of initial concentration and temperature

As illustrated in Fig. 3a, the adsorption capacity of Bif-MCM-41 for tetracycline increases consistently with the initial concentration (C_i , mg L^{-1}) factor across all three temperatures (T , K). This upward trend is prominent from 5 to 100 mg L^{-1} , after which the adsorption capacity begins to plateau. This can be explained by the availability of abundant active sites on the adsorbent surface at lower concentrations, allowing for more tetracycline molecules to be adsorbed. However, as the initial concentration increases, these active sites become increasingly occupied, leading to

Fig. 3 Depiction of **a** the effect of C_i on Q_e , **b** the impact of C_i on removal efficiency, **c** the relationship between C_e and Q_e , along with corresponding nonlinear isotherm model fits, **d** linear fitting curve for the Langmuir isotherm, **e** linear fitting curve for the Freundlich isotherm, and **f** linear fitting curve for the Temkin isotherm



saturation and a subsequent leveling off of the adsorption capacity rate.

In contrast, as shown in Fig. 3b, the removal efficiency diminishes progressively with higher C_i values. This inverse relationship occurs because, at elevated concentrations, the ratio of available adsorption sites to tetracycline molecules decreases, resulting in a reduction in the overall removal efficiency. Essentially, as the concentration of tetracycline increases, the fixed number of adsorption sites on the adsorbent is insufficient to maintain a high level of removal efficiency.

Temperature also exerts a significant influence on adsorption performance. Increasing the temperature has a detrimental effect on both adsorption capacity and removal efficiency at every concentration point. This negative effect suggests that the adsorption process is exothermic, where higher temperatures reduce the physical interaction strength between tetracycline molecules and the adsorbent surface. According to the literature, the heightened kinetic energy of molecules at elevated temperatures serves as the underlying cause for this phenomenon, as it promotes a more rapid desorption process and disrupts the

adsorbate-adsorbent interaction (Bhatnagar and Sillanpää 2010).

These experimental results provide critical insights for analyzing the isothermal and thermodynamic parameters of tetracycline adsorption on Bif-MCM-41. To evaluate these parameters, three two-parameter isotherm models—Langmuir (Langmuir 1916, 1918), Freundlich (1907), and Temkin (1940)—along with the three-parameter Redlich–Peterson model (Redlich and Peterson 1959), were applied. Both nonlinear and linear fitting methods were employed (Fig. 3c–f), with the fitting results at three different temperatures summarized in Table 1. (The parameters associated with the nonlinear and linear fitting models are denoted by (NL) and (L), respectively, as presented in Table 1.)

The nonlinear fitting approach is generally considered superior to linear fitting for several reasons. Nonlinear fitting avoids the distortions and biases introduced by linearization, which can alter the true relationship between the adsorption variables. Moreover, it offers a more precise fit by directly applying the original adsorption equations to the experimental data, resulting in higher correlation coefficients and more accurate estimation of isotherm parameters. Considering the R^2_{adj} values, the nonlinear fitting of the adsorption models at all three temperatures reveals the

following trend: Langmuir > Temkin > Freundlich. In contrast, the linear fitting models exhibit a slightly different order: Langmuir > Freundlich > Temkin.

When comparing the maximum adsorption capacities estimated by the Langmuir model ($Q_{m,cal}$, mg g⁻¹) through nonlinear fitting with the experimental maximum adsorption capacities ($Q_{m,exp}$, mg g⁻¹), a relatively close alignment is observed across the temperatures: $T=293$ K ($Q_{m,cal}=765.4$, $Q_{m,exp}=734.4$), $T=303$ K ($Q_{m,cal}=751.3$, $Q_{m,exp}=687.2$), and $T=313$ K ($Q_{m,cal}=708.3$, $Q_{m,exp}=638.4$). Notably, the linear fitting of the Langmuir model produced an even closer correspondence with the experimental values: $T=293$ K ($Q_{m,cal}=714.3$, $Q_{m,exp}=734.4$), $T=303$ K ($Q_{m,cal}=666.7$, $Q_{m,exp}=687.2$), and $T=313$ K ($Q_{m,cal}=625.0$, $Q_{m,exp}=638.4$).

The reduction in the Freundlich model parameters n (dimensionless) and K_F ((mg g⁻¹)(L mg⁻¹)^{1/n}) with increasing temperature, observed in both linear and nonlinear fittings, typically suggests a decline in adsorption intensity and capacity. This trend is consistent with the notion that higher temperatures may weaken the interactions between the adsorbent and adsorbate, making the adsorption process less favorable. The literature supports the notion that elevated temperatures can diminish binding affinity, particularly by weakening physical interactions like π - π stacking and hydrogen bonding. This reduction in interaction strength is reflected in the current study, where the functional groups of the Bif-MCM-41 adsorbent and tetracycline exhibit decreased adsorption efficiency as temperature increases.

Similarly, in the Temkin model, the decrease in B_T (J mol⁻¹) and K_T (L mg⁻¹) parameters as the temperature rises reflects a reduction in the energy of adsorption. This implies that at higher temperatures, the adsorbate-adsorbent interaction becomes weaker, thereby lowering the adsorption capacity. This trend is consistent with the exothermic nature of the adsorption process, where increasing the temperature typically results in decreased adsorption due to the enhanced kinetic energy of the molecules, which favors desorption.

For the RP model, a decrease in the K_{RP} (L g⁻¹) and α_{RP} (L mg⁻¹) parameters with increasing temperature suggests a reduction in the effectiveness of the adsorption process at higher temperatures. This could be attributed to the increased desorption rates or a decline in the affinity between tetracycline molecules and Bif-MCM-41. The g parameter within the Redlich–Peterson model indicates the proximity of the adsorption behavior to either the Langmuir or Freundlich models. A g value closer to unity implies that the adsorption process is more closely aligned with the Langmuir model. Based on the obtained g values, it can be inferred that the Langmuir model more accurately represents the adsorption behavior of tetracycline on Bif-MCM-41 compared to the Freundlich model.

Table 1 Isotherm parameters and corresponding values after nonlinear and linear fitting analyses

Models	Parameters	$T=293$ K	$T=303$ K	$T=313$ K
Langmuir	$Q_{m,exp}$	734.4	687.2	638.4
	$Q_{m,cal}$	765.4 (NL) 714.3 (L)	751.3 (NL) 666.7 (L)	708.3 (NL) 625.0 (L)
	K_L	0.5532 (NL) 1.1666 (L)	0.2203 (NL) 0.2143 (L)	0.1623 (NL) 0.1379 (L)
	R^2_{adj}	0.9657 (NL) 0.9996 (L)	0.9646 (NL) 0.9972 (L)	0.9636 (NL) 0.9941 (L)
	n	4.780 (NL) 3.134 (L)	3.760 (NL) 2.495 (L)	3.499 (NL) 2.350 (L)
	K_F	309.5 (NL) 229.4 (L)	221.4 (NL) 142.4 (L)	185.5 (NL) 114.9 (L)
Freundlich	R^2_{adj}	0.8533 (NL) 0.9373 (L)	0.8166 (NL) 0.9193 (L)	0.8273 (NL) 0.9238 (L)
	B_T	0.0119 (NL) 28.916 (L)	0.0102 (NL) 25.584 (L)	0.0101 (NL) 26.175 (L)
	K_T	84.029 (NL) 84.015 (L)	12.329 (NL) 12.329 (L)	6.884 (NL) 6.884 (L)
	R^2_{adj}	0.9058 (NL) 0.9058 (L)	0.8417 (NL) 0.8417 (L)	0.8555 (NL) 0.8555 (L)
R-P	K_{RP}	463.2 (NL)	157.9 (NL)	110.8 (NL)
	α_{RP}	0.6114 (NL)	0.1917 (NL)	0.1444 (NL)
	g	0.9889 (NL)	0.9665 (NL)	0.9778 (NL)
	R^2_{adj}	0.9427 (NL)	0.9290 (NL)	0.9352 (NL)

NL nonlinear fitting model, L linear fitting model

The plot of $\ln K_e^o$ versus $1/T$ depicted in Fig. S2, along with the thermodynamic parameters (Table S1) derived from the corresponding linear equation, reveals insightful details about the adsorption process of tetracycline on Bif-MCM-41. $\Delta H_{\text{Ads.}}^o$ was calculated to be $-46.982 \text{ kJ mol}^{-1}$, while $\Delta S_{\text{Ads.}}^o$ was estimated at $0.057 \text{ kJ mol}^{-1} \text{ K}^{-1}$. These values suggest an exothermic and spontaneous adsorption process. The negative enthalpy indicates that the adsorption is driven by physical forces rather than chemical bonding, aligning with the expected behavior for physical adsorption where lower temperatures typically favor adsorption.

The decrease in $\Delta G_{\text{Ads.}}^o$ from $-30.281 \text{ kJ mol}^{-1}$ at 293 K to $-29.711 \text{ kJ mol}^{-1}$ at 303 K and further to $-29.141 \text{ kJ mol}^{-1}$ at 313 K suggests that the spontaneity of the adsorption process slightly diminishes as temperature increases. This trend is characteristic of physical adsorption, where elevated temperatures can reduce the overall adsorption efficiency due to the increased kinetic energy of the molecules, which tends to favor desorption. Moreover, the relatively low enthalpy change is consistent with adsorption mechanisms dominated by weak interactions such as π - π stacking and hydrogen bonding. In the context of this study, the functional groups ($-\text{NH}-$, $-\text{NH}_2$, phenyl) on the Bif-MCM-41 adsorbent likely engage in π - π stacking with the aromatic rings of tetracycline and hydrogen bonding with its hydroxyl, ketone, and amino groups. These interactions are typically sensitive to temperature changes, and the observed thermodynamic data reinforce the notion that the adsorption of tetracycline onto Bif-MCM-41 is predominantly governed by these physical interactions. As temperature increases, the weakening of these interactions could explain the slight reduction in the overall adsorption capacity, as indicated by the Gibbs free energy values.

The influence of time

Figure 4a demonstrates how contact time (t , min) influences both the adsorption capacity and removal efficiency of tetracycline onto Bif-MCM-41 over a 120 min duration at 293 K. Initially, as contact time extends from 5 to 30 min, there is a sharp rise in both metrics, which then progresses at a reduced rate until stabilizing at approximately 50 min. To elucidate the adsorption kinetics, three well-established kinetic models—PFO (Ls 1898), PSO (Yuh-Shan 1995), and Elovich (1939)—were applied in both nonlinear and linear forms. Table 2 presents the estimated data from these models, while Fig. 4b illustrates the experimental adsorption capacities at any time t (Q_t , mg g^{-1}) values at 293 K versus time, alongside the nonlinear fits for PFO, PSO, and Elovich models. The linear fittings of these models are depicted in Fig. 4c–e.

According to Table 2, the nonlinear fitting analysis reveals the best model fit in the following order:

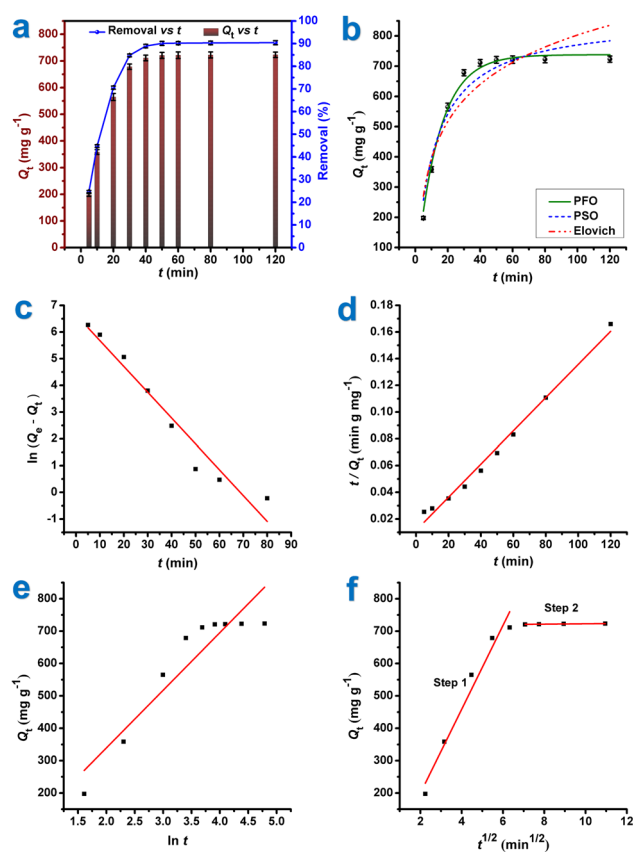


Fig. 4 **a** The relationship between $Q_{e,cal.}$ and time at 293 K. **b** nonlinear kinetic models fitted to the experimental data at 293 K. The linear fitting curves for **c** PFO model, **d** PSO model, **e** Elovich model, and **f** IPD kinetic models

PFO > PSO > Elovich. Conversely, the linear model fitting ranks as PSO > PFO > Elovich.

To determine the most accurate representation of the adsorption kinetics, the calculated equilibrium adsorption capacities ($Q_{e,cal.}$, mg g^{-1}) at 293 K from these models were compared to the experimental equilibrium adsorption capacity ($Q_{e,exp.} = 723.2 \text{ mg g}^{-1}$). The comparison highlights that the $Q_{e,cal.}$ values from the PFO model are closer to $Q_{e,exp.}$ than those from the PSO model after both linear and nonlinear fitting methods. Additionally, for the PFO model, the nonlinear fitting-derived $Q_{e,cal.}$ ($Q_{e,cal.} = 737.9 \text{ mg g}^{-1}$) is more consistent with the linear fitting-derived $Q_{e,cal.}$ ($Q_{e,cal.} = 769.3 \text{ mg g}^{-1}$), indicating superior accuracy of the nonlinear PFO model in depicting tetracycline adsorption on Bif-MCM-41.

The IPD model further elucidates the adsorption process, revealing a two-step mechanism as shown in Fig. 4f. This two-step behavior aligns with literature on adsorption phenomena in mesoporous silica materials, where, in accordance with established theories, the adsorption process typically initiates with a comparatively swift accumulation of

Table 2 Kinetic parameters and corresponding values after nonlinear and linear fitting analyses for adsorption of tetracycline on Bif-MCM-41 from water

Models	Parameters	Values
PFO	$Q_{e,cal.}$	737.9 (NL) 769.3 (L)
	k_1	0.071 (NL) 0.097 (L)
	$R^2_{adj.}$	0.9911 (NL) 0.9477 (L)
PSO	$Q_{e,cal.}$	860.9 (NL) 833.3 (L)
	k_2	9.9×10^{-5} (NL) 12.6×10^{-5} (L)
	$R^2_{adj.}$	0.9383 (NL) 0.9982 (L)
Elovich	α	161.95 (NL) 162.48 (L)
	β	0.0056 (NL) 0.0056 (L)
	$R^2_{adj.}$	0.8553 (NL) 0.8553 (L)
IPD	C_{IPD1}	−59.63 (Step 1)
	C_{IPD2}	716.88 (Step 2)
	$k_{diff.1}$	129.59 (Step 1)
	$k_{diff.2}$	0.59 (Step 2)
	$R^2_{adj.1}$	0.9562 (Step 1)
	$R^2_{adj.2}$	0.9225 (Step 2)

adsorbate molecules on the external surface, subsequently transitioning into a more gradual diffusion phase within the mesoporous structure. This finding confirms that both surface adsorption and intra-particle diffusion are integral to the tetracycline adsorption using Bif-MCM-41.

Regeneration and reusability of Bif-MCM-41

To evaluate the reusability of Bif-MCM-41 as an effective adsorbent for tetracycline, regeneration experiments were carried out for eight consecutive adsorption–desorption cycles. Each adsorption step was performed under optimized conditions. Due to the surface functionalization of Bif-MCM-41 with silane coupling agents bearing amine and phenyl groups, desorption was achieved using a polar organic solvent system capable of disrupting hydrogen bonding and π – π interactions.

The regeneration performance of Bif-MCM-41 over eight adsorption–desorption cycles is shown in Fig. 5.

In the first cycle, the removal efficiency of tetracycline was approximately 90%, which was considered as the reference value representing 100% adsorption capacity. Over the

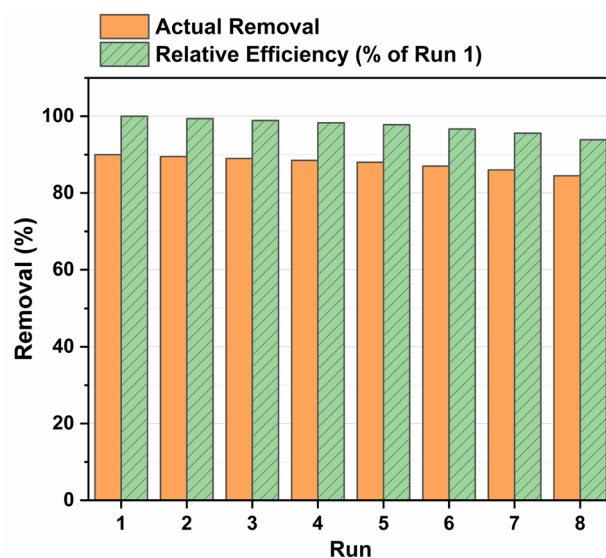


Fig. 5 Reusability performance of Bif-MCM-41 over eight successive adsorption–desorption cycles for tetracycline removal. The first run was considered as 100% adsorption efficiency (relative efficiency), and subsequent runs were evaluated relative to the initial performance to illustrate the gradual decrease in removal (%) over time

course of eight consecutive cycles, the removal percentages slightly decreased as shown in Fig. 5.

These results demonstrate that Bif-MCM-41 maintained a high adsorption performance even after eight cycles, with only a 6.1% decrease in adsorption efficiency compared to the initial cycle. The gradual decline may be attributed to the incomplete desorption of TC molecules or slight structural changes in the adsorbent surface after repeated washing and drying steps.

The effective regeneration with minimal loss in adsorption capacity confirms the high stability and reusability of Bif-MCM-41, making it a promising candidate for practical applications in environmental remediation of antibiotic contaminants.

Comparative study

Based on the data provided in Table 3, a comparison of the maximum adsorption capacity (Q_m) of different adsorbents in optimum conditions for tetracycline reveals that Bif-MCM-41 demonstrates the highest adsorption capacity among all the adsorbents listed. Here is a detailed comparison:

Maximum adsorption capacity (Q_m , mg g^{−1})

Bif-MCM-41 demonstrates a notably high adsorption capacity with a $Q_m = 734.4$ mg g^{−1}, significantly outperforming all other compared adsorbents. The next highest capacity

Table 3 Comparison with other adsorbents for tetracycline removal (RT: room temperature)

Adsorbent	Q_m (mg g ⁻¹)	pH	T (K)	T (min)	Year	References
Bif-MCM-41	734.4	7.0	293	50	2025	This work
	687.2	7.0	303	50		
	638.4	7.0	313	50		
AZN	454.55	6.7	298	20	2023	Al-Salihi et al. (2023)
ZnO/SA-NFs	248.6	3.0	—	120	2023	Zhang et al. (2023)
(PVA-CA) gel beads	231.43	5.0	298	—	2022	Liao et al. (2022)
CA-CNTs	230.2	3.5	318	2000	2022	Zhang et al. (2022)
Shrimp shell waste	~230	7.0	328	864	2020	Chang et al. (2020)
CTM	215.31	7.0	298	>45	2020	Ahamad et al. (2020)
Fe ₃ O ₄						
CA-BT	153.89	7.0	318	1200	2019	Zhang et al. (2019)
Alg-Fe ₃ O ₄	150	10	298	20	2022	Rahmani et al. (2022)
Fe ₃ O ₄	92.01	7.0	298	360	2022	Sheng et al. (2022)
PDA-ZIF-8						
NiFe ₂ O ₄	50	7.0	298	—	2023	Alessandro Idehara et al. (2023)
CoFe ₂ O ₄	43	7.0	298	—		
CD-MCP	35	10.0	RT	720	2016	Zhang et al. (2016)
Olive pomace	1.6	8.0	298	120	2019	Rizzi et al. (2019)
Chitosan/Olive pomace film	16	8.0	298	120		

is observed for AZN at 454.55 mg g⁻¹ (Al-Salihi et al. 2023), followed by ZnO/SA-NFs (Zhang et al. 2023) with 248.6 mg g⁻¹. In contrast, other materials such as NiFe₂O₄ and CoFe₂O₄ (Alessandro Idehara et al. 2023) as well as CA-CNTs (Zhang et al. 2022) and Fe₃O₄@PDA-ZIF-8 (Sheng et al. 2022) exhibit much lower adsorption capacities, ranging between 50 and 230.2 mg g⁻¹.

pH of adsorption

Bif-MCM-41 operates effectively at a neutral pH of 7.0, which aligns with the optimal pH range reported for many efficient adsorbents. Similarly, Fe₃O₄@PDA-ZIF-8 and CTM@Fe₃O₄ (Ahamad et al. 2020) also perform at pH 7.0, although their adsorption capacities are considerably lower, recorded at 92.01–230.2 mg g⁻¹ and 215.31–230.2 mg g⁻¹, respectively. Some materials demonstrate optimal performance under acidic conditions (e.g., ZnO/SA-NFs at pH 3.0 (Zhang et al. 2023)) or basic environments (e.g., Alg-Fe₃O₄ at pH 10.0 (Rahmani et al. 2022)); however, these extremes do not yield adsorption capacities superior to that of Bif-MCM-41.

Adsorption time

The Bif-MCM-41 adsorbent achieved its maximum adsorption capacity within a relatively short contact time of 50 min. Although AZN and Alg-Fe₃O₄ also demonstrate rapid adsorption kinetics, reaching equilibrium in 20 min, their overall adsorption capacities are markedly lower. In contrast,

certain adsorbents such as Fe₃O₄@PDA-ZIF-8 and CD-MCP (Zhang et al. 2016) require considerably longer contact times of 360 min and 720 min, respectively, yet still exhibit reduced adsorption performance compared to Bif-MCM-41.

Bif-MCM-41 not only outperforms all other adsorbents in terms of adsorption capacity but also shows an optimal balance between other parameters such as pH and time. This makes it the most effective adsorbent for tetracycline removal under the conditions studied, as compared to the other materials reported in Table 3. Synthesized Bif-MCM-41 offers a distinct advantage by utilizing silica extracted from sorghum, a sustainable and bioresource-derived material. The synthesis process is environmentally friendly, as it avoids the use of toxic chemicals, aligning with green chemistry principles. This innovative approach not only promotes sustainability but also contributes to the development of effective, eco-friendly solutions for adsorption applications. Bif-MCM-41's smart design incorporates phenol and amine groups as adsorptive sites inside its pores, improving tetracycline adsorption efficiency and offering a sustainable remediation solution.

Economic and practical considerations for Bif-MCM-41

The economic viability and practical applicability of adsorbent materials are essential for their transition from laboratory-scale development to real-world implementation. The synthesis approach adopted for Bif-MCM-41 in this study was deliberately designed to be cost-effective, sustainable, and scalable.

A key economic advantage lies in the use of sedge (*Carex riparia*), a renewable and underutilized agricultural residue, as the silica precursor. Unlike conventional silica sources that often involve high-purity reagents or energy-intensive processing, phytogenic silica extracted from sedge offers a low-cost and environmentally friendly alternative. Sedge grows abundantly in wetland ecosystems and requires no cultivation inputs, thereby reducing raw material costs and supporting circular economy principles through biomass valorization.

Furthermore, the one-pot synthesis route employed in this work presents substantial practical benefits over conventional multi-step post-grafting methods. By integrating the functionalization process within the initial synthesis stage, this method eliminates the need for additional solvents, surface activation treatments, and extended reaction times typically associated with post-synthesis modification. As a result, the one-pot method reduces both chemical consumption and energy input, improving overall process efficiency and reducing operational costs.

The use of commercially available silane coupling agents—AEAE-APTMS and BDPP-APTES—under mild reaction conditions (ambient pressure and moderate temperatures) further enhances the process's scalability. These reagents are widely accessible and compatible with green chemistry protocols, making them suitable for industrial-scale production. Additionally, the elimination of toxic solvents and the use of water as the primary reaction medium contribute to the environmental compatibility and safety of the synthesis process.

Overall, the synthesis of Bif-MCM-41 aligns well with the principles of green chemistry and sustainable engineering. The combination of a renewable silica source, solvent-free functionalization, and an energy-efficient one-pot synthesis strategy offers a scalable and economically attractive pathway for producing high-performance adsorbents suitable for environmental remediation, particularly in water treatment applications targeting pharmaceutical contaminants such as tetracycline.

Conclusion

In this study, a novel bi-functionalized ordered mesoporous silica (Bif-MCM-41) was synthesized using sedge-derived phytogenic silica—an underutilized, renewable agricultural waste—via an environmentally friendly one-pot method. This approach not only simplifies the synthesis process by integrating mesostructure formation and surface functionalization in a single step but also eliminates the need for solvent- and energy-intensive post-grafting procedures,

aligning with green chemistry and circular economy principles (Shirazian et al. 2025b).

The incorporation of two silane coupling agents, AEAE-APTMS and BDPP-APTES, introduced amine and phenyl groups onto the silica surface, enabling dual-mode adsorption through hydrogen bonding and π - π stacking interactions. Structural characterization confirmed the successful formation of an ordered mesoporous structure with a uniform Bacillus-like or rod-shaped morphology, as observed by FESEM and TEM, and a high BET surface area of $988 \text{ m}^2 \text{ g}^{-1}$. A bimodal pore distribution—dominated by micropores—was revealed by 2D-NLDFT analysis, contributing to efficient mass transfer and increased adsorption capacity.

Adsorption studies demonstrated that the Langmuir isotherm best described the equilibrium data, suggesting monolayer coverage a maximum adsorption capacity of 765.4 mg g^{-1} at 293 K, while kinetic modeling indicated that the process followed pseudo-first-order behavior. The intra-particle diffusion model supported a two-step mechanism involving rapid surface adsorption followed by slower diffusion into pores. Thermodynamic analysis confirmed the process was spontaneous and exothermic, driven primarily by physical interactions. Additionally, reusability studies showed that Bif-MCM-41 retained over 85% of its initial adsorption capacity after five cycles, highlighting its practical stability.

This work presents a comprehensive evaluation of Bif-MCM-41 for the adsorption of tetracycline, combining both linear and nonlinear modeling, thermodynamic assessments, and regeneration analysis. The combination of a sustainable precursor, eco-friendly synthesis route, and superior adsorption performance demonstrates the high potential of Bif-MCM-41 for advanced water treatment applications, particularly in the removal of pharmaceutical contaminants.

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project administration, visualization, writing—original draft, and writing—review and editing.

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Declarations

Ethical standard This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent In this article, no patient care was involved.

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